

David C. Lay: Linear Algebra and its Applications

Exercise 2.8

5. Let $v_1 = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}$, $v_2 = \begin{bmatrix} -4 \\ -5 \\ 8 \end{bmatrix}$ and $w = \begin{bmatrix} 8 \\ 2 \\ -9 \end{bmatrix}$. Determine if w is in the subspace of $\mathbb{R}^3$ generated by v_1 and v_2 .

Solution: By Example 1, $\text{Span}\{v_1, v_2\}$ is a subspace of $\mathbb{R}^3$.

By Defn, it is the set of all the linear combinations of v_1 and v_2 . We need to verify that whether $c_1 v_1 + c_2 v_2 = w$ for some $c_1, c_2 \in \mathbb{R}$.

By Defn. of Ax , $[v_1 \ v_2] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = w$.

By Theorem 3 from Chapter 1, the above matrix equation has the same solution set as the system of equations corresponding to the augmented matrix $[v_1 \ v_2 \ w]$.

$$\begin{bmatrix} 2 & -4 & 8 \\ 3 & -5 & 2 \\ -5 & 8 & -9 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row 1 = Row 1}} \begin{bmatrix} 1 & -2 & 4 \\ 3 & -5 & 2 \\ -5 & 8 & -9 \end{bmatrix} \xrightarrow[-3 \times \text{Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -10 \\ -5 & 8 & -9 \end{bmatrix} \xrightarrow[\text{Row 3} + 5 \times \text{Row 1}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -10 \\ 0 & -2 & 11 \end{bmatrix} \xrightarrow[+2 \times \text{Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -10 \\ 0 & 0 & -9 \end{bmatrix}$$

It's very easy to verify that the final matrix is in echelon form.

Row 3 of the matrix is of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$.
 $\therefore$ By Theorem 2 from Chapter 1, the matrix equation $[v_1 \ v_2] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = w$ is inconsistent, i.e. no solution set exists.

$\therefore w \notin \text{Span}\{v_1, v_2\}$ $\square$

7. Let $v_1 = \begin{bmatrix} 2 \\ -8 \\ 6 \end{bmatrix}$, $v_2 = \begin{bmatrix} -3 \\ 8 \\ -7 \end{bmatrix}$, $v_3 = \begin{bmatrix} -4 \\ 6 \\ -7 \end{bmatrix}$, $p = \begin{bmatrix} 6 \\ -10 \\ 11 \end{bmatrix}$ and $A = [v_1 \ v_2 \ v_3]$

- How many vectors are in $\{v_1, v_2, v_3\}$?
- How many vectors are in $\text{Col } A$?
- Is p in $\text{Col } A$? Why or why not?